MIL-D-17984A(ASG)
1 MAY 1956
Superseding
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MILITARY SPECIFICATION

DATA PRESENTATION REQUIREMENTS, INSTALLED ENGINE PERFORMANCE AND AIR INDUCTION SYSTEMS

This specification has been approved by the Department of the Air Force and by the Navy Bureau of Aeronautics.

1. SCOPE

1.1 This specification covers requirements for presentation of performance characteristics of power plant installations and inlet geometry systems, as follows, for all ramjet, pulse jet, turbojet, turboprop, and other air-breathing-type propulsion systems other than reciprocating engines for all piloted aircraft, guided missiles, and target drones:

(a) The basic engine performance data, corrected for installation effects, to be presented as the source of the values of power available in calculating aircraft performance under the Navy requirements of Specification MIL-D-8706 and Bureau of Aeronautics publication SAR-304, or Air Force requirements of Specifications MIL-D-8034 and MIL-D-7579.

(b) The desired method of calculating and presenting these installation losses.

(c) The information required on the control systems used in manipulating variable geometry features of power plant installations.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and publication, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-8629	Airplane Strength and Rigidity
MIL-D-7579	Data, Development Airplane Engineering
MIL-D-8034	Data for Evaluation of Aircraft Proposals
MIL-D-8706	Data, Design: Contract Requirements for Aircraft
MIL-S-5701	Structural Criteria, Piloted Airplanes, Definitions and Symbols

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PUBLICATIONS

Bureau of Aeronautics

SAR-304 General Design Information Requirements for Informal Aircraft Proposals

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The calculation of aircraft performance requires an accurate calculation of the power plant installation losses to determine the net thrust or shaft horsepower, as applicable, and the corresponding fuel consumption values of the engine as installed in the aircraft. In supersonic aircraft, when two-position or infinitely variable inlets are used to achieve low losses over a wide range of flight conditions, it is necessary that the control of the variable geometry features have an accuracy of regulation consistent with the demanded steady-state performance and have dynamic characteristics compatible with the dynamics of the basic power plant. Analyses submitted should be consistent with and reflect the best information available at the time of submittal of the reports or respective revisions.

3.1.1 The desired method of calculating these installation losses and presenting the analyses are presented below. They shall be used unless another well presented detailed engineering analysis of the problem, proposed by the contractor, is acceptable to the procuring activity.

3.2 Estimates of pressure recovery, friction factors, and any other empirical factors used in presenting the calculated data required by this specification, shall be justified and listed as to their source and accuracy.

3.3 The data required under this specification shall be submitted as a report, separate from the data of Specification MIL-D-8706 and Bureau of Aeronautics publication SAR-304, or Specifications MIL-D-8034 and MIL-B-7579. The required data shall be submitted in a report with the preliminary proposal, with revisions and addendums to be submitted at the conclusion of the applicable wind-tunnel test programs and at the conclusion of the applicable flight test program.

3.4 The basic engine performance data for ramjet, pulse jet, turbojet, turboprop, and other air-breathing-type propulsion systems other than reciprocating engine installations, corrected for installation effects, (inlet duct loss, inlet loss, exhaust duct loss, exhaust ejector loss, horsepower extraction loss, air bleed loss, cooling air drag, incremental scoop or additive drag, and drag owing to boundary layer removal) shall be as delineated herein.

3.4.1 Inlet and inlet duct analysis.-

3.4.1.1 A plan view and side view of the inlet and inlet duct providing pertinent dimensions shall be presented. In the case of variable area inlets, variable ramp angles, auxiliary inlets, or any other particular complexity of the inlet or duct, sufficient detail views shall be provided to enable the viewer to understand the nature of the complexity.

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3.4.1.2 A plot of variation of duct cross-sectional area and equivalent conical diffuser angle versus the duct station from the aircraft inlet to the engine inlet shall be presented. In the case of variable geometry inlets, the variation for various operating conditions shall be shown (see 3.5).

3.4.1.3 A plot of local velocity and dynamic pressure versus the duct station for maximum engine power or thrust, at a standard sea level static condition shall be presented.

3.4.1.4 Analysis of inlet duct.-

3.4.1.4.1 A tabulated analysis of the inlet duct shall be presented, as shown in table I.

TABLE I

Analysis of inlet duct

Section	$\frac{\Delta P_t}{q_1}$	$\frac{q_1}{q_e}$	$\frac{\Delta P_t}{q_e}$
0-1			
1-2			
2-3			
3-4			
4-5			
		Total	$\frac{\Delta P_{td}}{q_e}$

Where: ΔP_t = Loss in total pressure in section considered.

ΔP_{td} = Inlet duct total pressure loss.

q_1 = Dynamic pressure at discharge of section considered

q_e = Dynamic pressure at engine inlet.

3.4.1.4.1.1 In general, it may be assumed that $\Delta P_{td}/q_e$ will remain constant throughout the operating range where a constant duct and inlet configuration is used; therefore, calculations for maximum engine power or thrust at standard sea level static conditions generally will be satisfactory for all other conditions. In extreme cases and in cases where the inlet and duct configuration is variable, an analysis of the duct losses for each configuration of the inlet and duct which may be used is required. The engine inlet area shall be taken as the actual inlet area at the inlet face of the engine. In the case of centrifugal compressors, this shall be the inlet eye of the compressor, and in the case of axial flow compressors, this shall be the inlet annulus of the compressor.

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3.4.1.5 Analysis of duct inlet.-

3.4.1.5.1 A tabulated analysis of the duct inlet shall be presented, as shown in table II. For missiles or special purpose aircraft, typical important conditions in the mission or trajectory should be substituted as appropriate.

3.4.1.6 Plots shall be presented of mass flow ratio and capture area ratio versus corrected engine airflow, using free stream Mach number as a parameter. The variation of capture area ratio with angle of attack, up to the maximum angle practicably attainable by the aircraft, shall also be shown.

3.4.1.7 A plot shall be presented of total pressure recovery of the inlet and duct versus mass flow ratio at zero angle of attack, using free stream Mach number as a parameter.

3.4.1.8 A plot shall be presented of total pressure recovery of the inlet and duct versus angle of attack at design mass flow ratios, using free stream Mach number as a parameter. Practical negative angles of attack shall be included.

3.4.1.8.1 Plots shall be presented showing the radial and circumferential total pressure recovery variations obtained from model or full-scale tests for combinations of conditions shown below:

Mo = 0, Subsonic, 1.5, 2.0, ,max attainable by the aircraft. Mach numbers at which marked changes in the inlet operating characteristics occur shall be included.

α = Five representative angles of attack ranging from the minimum to maximum practicably attainable by the aircraft.

Ao/Ac = Approximately six representative values between the maximum and minimum normally to be encountered with the aircraft.

Ψ = Two representative angles of yaw, including the maximum likely to be encountered by the aircraft.

Where: - Mo = Free stream Mach number.

α = Airplane angle of attack.

Ao/Ac = Inlet capture area ratio.

Ψ = Aircraft yaw angle.

3.4.1.9 A plot shall be presented of total pressure recovery of the inlet and duct versus engine airflow, using engine power setting and inlet area as parameters. This plot shall be presented for each of the operating conditions shown in table II. The inlet-engine matching points selected and unstable operating regions shall be shown on this plot. The effects of yaw on inlet stable operating regions at angles up to the maximum expected for the aircraft shall also be shown on this plot.

3.4.2 Exhaust duct and exhaust ejector analysis.- This paragraph applies, as appropriate, only when the exhaust duct, nozzle, or ejector is supplied by the airframe manufacturer and when the effects of these parts are not included in the engine performance curves of the applicable engine specification.

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TABLE II

Analysis of duct inlet

Condition 1/	Mo	$\frac{Ao}{Ai}$	$\frac{Ao}{Ac}$	α	$\frac{\Delta P_{ti}}{qi}$	$\frac{\Delta P_{ti}}{qe}$	$\frac{\Delta P_{t2}}{qe}$	$\frac{\Delta P_{t2}}{to}$
1. Sea level - static (military rpm).								
2. Sea level - max power at takeoff speed at basic mission takeoff weight 2/.								
3. Sea level - max power clean aircraft Vmax at basic mission combat weight 2/.								
4. 15,000 ft - max power clean aircraft, at max R/C speed at basic mission combat weight 2/.								
5. Max cruise altitude attained during basic mission 3/, at speed and power for clean aircraft cruise.								
6. 35,000 ft - max power Vmax at basic mission combat weight 2/ with airplane in clean configuration.								
7. Max combat ceiling - Vmax at basic mission combat weight 2/ in clean configuration.								

1/ Where the inlet is of the variable type, the configuration to be used under each of the above conditions shall be considered in the analysis.

2/ As defined by Specification MIL-S-5701 for Air Force aircraft, or Specification MIL-A-8629 for Naval aircraft.

3/ As defined by the detail specification.

Where: Mo = Free stream Mach number.

$\frac{Ao}{Ai}$ = Mass flow ratio = $\frac{\text{Free stream tube area}}{\text{Inlet area}}$

$\frac{Ao}{Ac}$ = Capture area ratio = $\frac{\text{Free stream tube area}}{\text{Inlet capture area}}$

α = Angle of attack of aircraft.

ΔP_{ti} = Duct inlet total pressure loss.

qi = Dynamic pressure at duct inlet.

qe = Dynamic pressure at engine inlet.

ΔP_{t2} = Over-all total pressure loss of inlet and duct; $\Delta P_{t2} = \Delta P_{ti} + \Delta P_{td}$.

P_{to} = Absolute free stream total pressure.

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3.4.2.1 A plan view, and side view if not axially symmetric of the exhaust duct and exhaust outlet, shall be presented. In the case of variable area exhaust outlets, or any other particular complexity of the exhaust duct or outlet, sufficient detail views shall be shown, to enable the viewer to understand the nature of the complexity.

3.4.2.2 A plot shall be presented of variation of duct cross-sectional area and diffuser or nozzle angles, versus the duct station from the turbine to the outlet.

3.4.2.3 A plot shall be presented of exhaust duct Mach number versus the duct station from maximum engine power or thrust at standard sea level static conditions.

3.4.2.4 Analysis of exhaust duct.-

3.4.2.4.1 A tabulated analysis of the exhaust duct shall be presented, as shown in table III.

TABLE III

Analysis of exhaust duct

Section	$\frac{\Delta P_t}{P_{t1}}$	$\frac{\Delta P_{t1}}{P_{t5}}$	$\frac{\Delta P_{t5}}{P_{t5}}$
0-1			
1-2			
2-3			
3-4			
4-5			
		Total $\frac{\Delta P_{t5}}{P_{t5}}$	

Where: ΔP_t = Loss in total pressure in section considered.

P_{t1} = Total pressure at discharge of section considered.

P_{t5} = Total pressure at turbine exit.

ΔP_{t5} = Total loss in turbine exit total pressure in the exhaust duct.

3.4.2.4.1.1 In cases where the exhaust configuration is variable and furnished by the airframe manufacturer, an analysis of the duct losses for each configuration of the duct and exit which may be used shall be necessary. This analysis shall be made for each of the conditions shown in table II at each exit configuration, as stated above.

3.4.2.5 In installations where an exhaust ejector is used, a complete description of the ejector shall be presented. This shall include spacing ratio and diameter ratio (see 6.2.1), primary and secondary mass flow, and variations thereof, in cases of variable area exhaust nozzles. This shall also include inlet area and recovery for the secondary air inlet.

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3.4.2.5.1 Plots shall be presented showing the variation of ejector thrust (as a percentage of net thrust) with airspeed, altitude, and engine power setting. A sample calculation of the ejector thrust shall be presented.

3.4.3 Losses due to power extraction.-

3.4.3.1 All equipment using power extracted directly from the engine drive pads shall be listed, and a tabulation of the horsepower required shall be presented. If appropriate, a plot of horsepower extracted versus engine rpm shall be presented. Any significant changes in horsepower extraction during the mission shall be indicated.

3.4.4 Losses due to engine air bleed.-

3.4.4.1 All equipment and systems utilizing engine bleed air shall be listed, and a tabulation of the weight flow of air required shall be presented. If appropriate, plots of weight flow versus altitude at various Mach numbers shall be presented. The estimated continuous and maximum airflows, with approximate durations indicated, shall be shown on these plots.

3.4.5 Miscellaneous items affecting engine performance.-

3.4.5.1 The change in engine performance associated with any other installation characteristics shall be indicated. For example, the temperature effect of oil or generator cooling air discharging into the engine inlet.

3.4.6 Cooling air drag.-

3.4.6.1 The engine cooling air system, other than the secondary air ejector system, shall be discussed sufficiently as to the type used. Pertinent data, such as cooling air inlet recovery versus Mach number, inlet area, required cooling air mass flow, and exit angle shall be presented. A plot shall be presented of cooling air momentum drag versus altitude, using true airspeed as a parameter.

3.4.7 Analysis of supersonic inlet drag.-

3.4.7.1 Supersonic inlet drag (see 6.2.3) shall be determined. This drag is the summation of additive drag or scoop incremental drag, depending on the inlet location, and the thrust force due to inlet lip suction.

3.4.7.2 For supersonic aircraft with nose, leading edge, or pod inlets, a plot shall be presented of capture area ratio versus free stream Mach number, using the ratio of supersonic inlet drag coefficient to total drag coefficient as a parameter. Additive drag shall be used in the calculation of supersonic inlet drag.

3.4.7.3 For supersonic aircraft with inlets so located that a portion of the forebody is wetted by inlet air, it shall be stipulated whether or not the pressure drag of this wetted area has been included in the total drag of the aircraft body. If this drag has not been included, then a presentation, using additive drag, as specified in 3.4.7.2, shall be made. If the pressure drag of the wetted area has been included in the total drag, then a plot shall be presented, as in 3.4.7.2, but using scoop incremental drag, rather than additive drag, in the calculation of supersonic inlet drag.

3.4.7.4 A sample of the calculations used to obtain supersonic inlet drag shall be presented.

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3.4.8 Drag due to inlet boundary layer removal.-

3.4.8.1 The method of calculating boundary layer thickness at the inlet shall be presented. A plot of boundary layer thickness versus free stream Mach number, using constant altitude as a parameter, shall be shown.

3.4.8.2 The boundary layer removal system shall be discussed sufficiently as to type of removal used. Drag of the system shall be presented as a momentum drag.

3.4.9 Over-all effect upon engine installation performance of various installation losses.-

3.4.9.1 A tabulated analysis of the over-all effect shall be presented as shown in table IV for the same conditions as presented in table II.

3.4.9.2 A plot shall be presented of mass flow ratio versus free stream Mach number for maximum limiting engine power or thrust, showing a continuous envelope line of acceleration to Vmax at 35,000 feet, limiting velocity dive to sea level and acceleration to Vmax at sea level. The inlet buzz region, using the ratio of total pressure fluctuation increment to average total inlet pressure as a parameter, shall be shown on this plot.

3.4.9.3 A plot of net thrust and fuel flow, corrected for installation losses, versus airspeed at constant altitudes for at least five practical engine power settings shall be presented. These plots shall be repeated, as appropriate, for the afterburning condition. A copy or reproduction of the engine manufacturer's specification used shall also be included.

3.5 The information required on the control systems used in manipulating variable geometry features of power plant installations shall be as delineated below.

3.5.1 Variable geometry control system.-

3.5.1.1 A schematic drawing shall show all elements (mechanical, hydraulic, electrical, pneumatic aerodynamic, etc) which constitute the installation control system. A general description explaining the functioning of the complete system, including functions of individual elements and the relation of the system to the power plant shall accompany the schematic arrangement.

3.5.1.2 A reliability analysis report shall include assumed failure of each critical component in the most adverse position or condition, or both. Assumed failure of power supply to the control system also shall be included.

3.5.1.3 A design report shall be submitted, prior to or concurrently with the drawings specified in 3.5.1.1, showing methods of analysis of the stability and steady-state regulation characteristics of the control system in conjunction with the power plant during critical engine and airspeed transients. The report shall show the objective steady-state accuracy of regulation for the control system. The transfer function of the dynamic effect of duct geometry changes on the total pressure at the engine inlet shall be shown for each of the conditions in 3.4.1.5.1.

3.5.1.3.1 The description of the methods of analysis shall be sufficiently detailed to permit review. Derivation of equations, sources of parameter values, and sample calculations shall be included.

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TABLE IV

Over-all effect upon engine installation performance of various installation losses

Condition	$\frac{\Delta P_{t2}}{P_{to}}$	$\frac{\Delta P_{t5}}{P_{t5}}$	HP_x	W_{BL}	$\frac{\Delta SHP}{SHP}$	$\frac{\Delta F_n}{F_N}$	$\frac{\Delta W_f}{W_f}$	$\frac{F_e}{F_N}$	F_c	$\frac{\Delta F_n + F_c + D}{F_N} + \frac{F_e}{F_N}$
1. Sea level - static (military rpm).										
2. Sea level - max power at takeoff speed at basic mission takeoff weight.										
3. Sea level - Max power clean aircraft Vmax at basic mission combat weight.										
4. 15,000 ft - max power clean aircraft, at max R/C speed at basic mission combat weight.										
5. Max cruise altitude attained during basic mission at speed and power for clean aircraft cruise.										
6. 35,000 ft - max power Vmax at basic mission combat weight with aircraft in clean configuration.										
7. Combat ceiling - Vmax at basic mission combat weight in clean configuration.										

Where:

HP_x = Total horsepower extracted from engine through drive pads.

W_{BL} = Total bleed air taken from engine compressor.

Δ SHP = Loss in shaft horsepower owing to inlet, inlet and exhaust ducts, horsepower extraction, and bleed air.

SHP = Shaft horsepower of engine without losses.

ΔF_n = Loss in net thrust owing to inlet, inlet and exhaust ducts, horsepower extraction and bleed air.

F_N = Net thrust of engine without losses.

ΔW_f = Change in fuel consumption of engine owing to inlet, inlet duct, and bleed air.

W_f = Fuel consumption of engine without losses.

F_e/F_N = Percent gain or loss in engine net thrust owing to ejector.

F_c = Cooling airflow momentum drag.

D = Sum of additive or scoop incremental drag and boundary layer removal drag.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Not applicable to this specification.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 Intended use.- It is intended that analyses for power plant installation and inlet-geometry-system characteristics of all ramjet, pulse jet, turbojet, and turboprop installations on piloted aircraft, guided missiles, and target drones shall be presented in accordance with the requirements of this specification when performance estimates are submitted to the Navy under Specification MIL-D-8706 and Bureau of Aeronautics publication SAR-304, or to the Air Force under Specifications MIL-D-8034 and MIL-D-7579. (See 3.3.)

6.2 Definitions.-

6.2.1 Spacing ratio is defined as the ratio of the distance between the primary and secondary nozzle exits to the diameter of the primary nozzle. (See 3.4.2.5.)

6.2.2 Diameter ratio is defined as the ratio of the secondary nozzle diameter to the primary nozzle diameter. (See 3.4.2.5.)

6.2.3 Supersonic inlet drag is defined as the drag of the inlet at supersonic flight speeds due to operation at mass flow ratios other than unity. (See 3.4.7.)

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Custodians:
Navy - Bureau of Aeronautics
Air Force

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION MIL-M-38780(USAF)

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CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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